

Appendix 30.A

2G Cellular – GSM

The *Global System for Mobile communications* (GSM) was the most successful 2G cellular communication system. While its technology is mostly outdated, and the focus on speech (and some elementary data services) not aligned with modern applications, it is still instructive to study this system. Firstly, it is still operative in significant portions of the world, in particular where its comparatively low cost and focus on basic communications services is desired. Secondly, it is instructive to study the principle of a relatively simple cellular system, as it might help to understand principles that are less readily learned from the extremely complicated 4G/5G systems.

30.A.1 Historical Overview

The development of GSM started in 1982. The European Conference of Postal and Telecommunications Administrations (CEPT), predecessor of the *European Telecommunication Standards Institute* (ETSI), founded the *Groupe Speciale Mobile*, with the mandate to develop proposals for a pan-European digital mobile communication system. Two goals were supposed to be achieved:

- First, a better and more efficient technical solution for wireless communications – it had become evident at that time that digital systems would be superior in respect to user capacity, ease of use, and number of possible additional services compared with the then-prevalent analog systems.
- Second, a single standard was to be realized all over Europe, enabling roaming across borders. This was not possible before, as *incompatible* analog systems were employed in different countries.

In the following years, several companies developed proposals for such a system. These proposals covered almost all possible technical approaches in different technical areas. For *multiple access*, Time Division Multiple Access (TDMA), Frequency Division Multiple Access (FDMA), and Code Division Multiple Access (CDMA) were suggested. The proposed *modulation techniques* were Gaussian Minimum Shift Keying (GMSK), 4-Frequency Shift Keying (4-FSK), Quadrature Amplitude Modulation (QAM), and Adaptive Differential Pulse Modulation (ADPM). *Transmission rates* varied from 20 kbit/s to 8 Mbit/s. All of the proposed systems were tested both in field tests and with a channel simulator (in Paris in 1986). Apart from technical considerations, marketing and political arguments influenced the decision-making process. FDMA could not be employed, as it would have required antenna diversity at the User Equipment (UE). Even though the technical feasibility of this diversity had been proven by the Japanese digital system, increased antenna sizes did not make it a desired option. CDMA was ultimately excluded, because the necessary signal processing seemed to be too expensive at that time. Therefore, only a TDMA system could survive the selection process. However, the final TDMA system was not a proposal from a single company, but rather a compromise system was developed. The reasons for this were of a political and not technical nature: selecting the proposal of one company as the standard would have given this specific company a large competitive advantage. Specific details of the compromise system were developed by a – now permanent – committee over the following two years and served as the basis for systems implemented in Europe after 1992.

In the early 1990s, it was realized that GSM should have functionalities that had not been included in the original standard. Therefore, the so-called phase-2 specifications, which included these functions, were developed until 1995. Further enhancements, which include packet radio (General Packet Radio Service (GPRS), see Sec. 30.A.12 and the more efficient modulation of Enhanced Data rates for GSM Evolution (EDGE), were introduced even later. Because of these extensions GSM is often referred to as the *2.5th generation system*, as its functionalities are beyond those of a second-generation system, but did not enable all third-generation functionalities.

The success of GSM exceeded all expectations. Though it was originally developed as a European system, it became a worldwide cellular communication standard,¹ with a number of subscribers that approached 4.0 billion in 2010. Since that time, it has lost a lot of subscribers to 3G, 4G (LTE) and 5G (NR). However, in 2020 there were still more than a billion 2G subscribers in the world, about 20% of worldwide subscribers, with Africa, Middle East, and South and South/East Asia the biggest markets. GSM thus remains an important part of the wireless ecosystem.

GSM can use different sets of carrier frequencies. The original GSM system uses carrier frequencies around 900 MHz. GSM1800, which was also called Digital Cellular System at the 1800-MHz band (DCS1800), was added later to support the increasing numbers of subscribers. Its carrier frequencies are around 1,800 MHz, the total available bandwidth is roughly three times larger than the one around 900 MHz, and the maximal transmission power of UEs is reduced. Apart from this, GSM1800 is identical to the original GSM. Thus, signal processing, switching technology, etc. can be reused without changes. The higher carrier frequency, which implies a smaller path gain, and reduced transmission power reduce the sizes of the cells significantly. This fact, combined with the bigger available bandwidth, leads to a considerable increase in network capacity.

30.A.2 System Overview

A GSM system consists essentially of three parts – namely, the *Base Station Subsystem* (BSS), the *Network and Switching Subsystem* (NSS), and the *Operation Support System* (OSS).

30.A.2.1 Base Station Subsystem

The BSS consists of *Base Transceiver Stations* (BTSs) and the *Base Station Controllers* (BSCs) (see Figure 30.A.1). The BTS establishes and maintains the connection to the UEs within its cell. The interface between the UE and the BTS is the air interface, called the *Um-interface* in the GSM context. The BTS hosts, at a minimum, the antennas and the Radio Frequency (RF) hardware of a BS, as well as the software for multiple access. Several – or, rarely, one – BTSs are connected to one BSC; they are either colocated, or connected via optical fiber, landline, directional microwave radio links, or similar connections. The BSC has a control functionality. It is, among other things, responsible for HandOver (HO) between two BTSs that are connected to the same BSC. The interface between BTS and BSC is called the *Abis-interface*. Distribution of the functionalities between BTS and BSC may differ depending on the manufacturer. In most cases, one BSC is connected to several BTSs. Therefore, it is possible to increase the efficiency of implementation by shifting as much functionality as possible to the BSC. However, this implies increased signaling traffic on the link between the BTS and the BSC, which might be undesirable. In general, the BSS covers a large set of functionalities. It is responsible for channel assignment, maintenance of link quality and handover, power control, coding, and encryption.

30.A.2.2 Network and Switching Subsystem

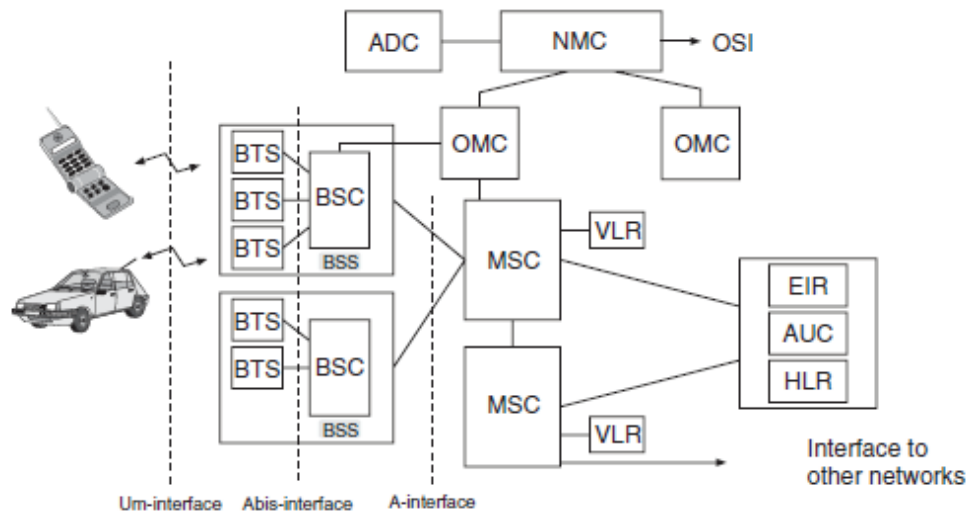
The main component of the NSS is the *Mobile-services Switching Center* (MSC), which controls the traffic between different BSCs (see Figure 30.A.1). One function of the MSC is *mobility management*, which comprises all the functions that are necessary to enable true mobility for subscribers. To give but one example, one function of the MSC is the management of HOs that occur when a UE is leaving the area of one BSC and moving into the area covered by another BSC. Other functions are the so-called *paging* and *location update*. All interactions with other networks – especially the landline *Public Switched Telephone Network* (PSTN) and the internet – are also performed by the MSC.

The NSS includes some databases, too. The *Home Location Register* (HLR) contains all the numbers of the mobile subscribers associated with one MSC and information about the location of each of these subscribers. In the event of an incoming call, the location of the desired subscriber is looked up in the HLR and the call is forwarded to this location.² Therefore, we can conclude that from time to time a traveling UE has to send updates of its location to its HLR. The *Visitor Location Register* (VLR) of one MSC contains all the

¹ Hence the reinterpretation of GSM from “Groupe Speciale Mobile” to “Global System for Mobile communications.”

² Actually the call is only forwarded to the BSC in whose area the subscriber is. Routing to and selection of one BTS is the responsibility of the BSC.

information about mobile subscribers from other HLRs that are in the area of this MSC and are allowed to roam in the network of this MSC. Furthermore, a temporary number will be assigned to the UE to enable the “host” MSC to establish a connection to the visiting UE.



30.A.2.3 Operating Support System

1. *Accounting*: how much does a specific call cost for a certain subscriber? There are also plenty of different services and features, from which each subscriber may choose an individual selection included in a specific plan. While this rich choice of services and prices is vital in the marketplace, the administrative support of this individualism is rather complicated. Examples are discussed in Section 30.A.10.
2. *Maintenance*: the full functionality of each component of the GSM network has to be maintained all the time. Malfunctions may either occur in the hardware or in the software components of the system. Hardware malfunctions are more costly, as they require a technician to drive to the location of the malfunction. In contrast, software is nowadays administrated from a central location. For example, new versions of switching software can be installed in the complete BSS from a central location, and activated all over the network at a specific time. Revision and maintenance software often constitutes a considerable part of the overall complexity of GSM control software.
3. *UE management*: even though all UEs have to pass a type approval, it may happen that “bad apple” devices, which cause systemwide interference, are operating in the network. These devices have to be identified and their further activities have to be blocked.
4. *Data collection*: the OSS collects data about the amount of traffic, as well as the quality of the links.

30.A.3 The Air Interface

GSM employs a combined FDMA/TDMA approach which further combines with Frequency Domain Duplexing (FDD) (see Chapter 18). Let us elaborate on these acronyms.

FDD

In the first GSM version, frequencies from 890 to 915 MHz and from 935 to 960 MHz were available. The lower band is used for the uplink (connection from the UE to the BS). The upper band is used for the downlink. The frequency spacing between the uplink and downlink for any given connection is 45 MHz. Therefore, relatively cheap duplex filters are sufficient for achieving very good separation between the uplink and downlink.

For GSM1800, the original frequency ranges are 1,710–1,785 MHz for the uplink, and 1,805–1,880 MHz for the downlink. The frequency bands used nowadays vary from country to country, and over time, as more and more bands are re-assigned to other cellular systems.

FDMA

Both uplink and downlink frequency bands are partitioned into a 200-kHz grid. The outer 100 kHz of each 25-MHz band are not used,³ as they are *guard bands* to limit interference in the adjacent spectrum, which is used by other systems. The remaining 124 200-kHz subbands are numbered consecutively by the so-called *Absolute Radio Frequency Channel Numbers* (ARFCNs).

TDMA

Due to the use of GMSK (see below) as modulation, each 200-kHz subband supports a data rate of 271 kbit/s. Each subband is shared by eight users. The time axis is partitioned into timeslots, which are periodically available to each of the possible eight users (Figure 30.A.2). Each timeslot is 576.92 μ s long, which is equivalent to 156.25 bits. A set of eight timeslots is called a *frame*; it has a duration of 4.615 ms. Within each frame, the timeslots are numbered from 0 to 7. Each subscriber periodically accesses one specific timeslot in every frame on one frequency subband. The combination of timeslot number and frequency band is called the *physical channel*. The kind of data that are transmitted over one such physical channel depends on the *logical channel* (see also Section 30.A.4).

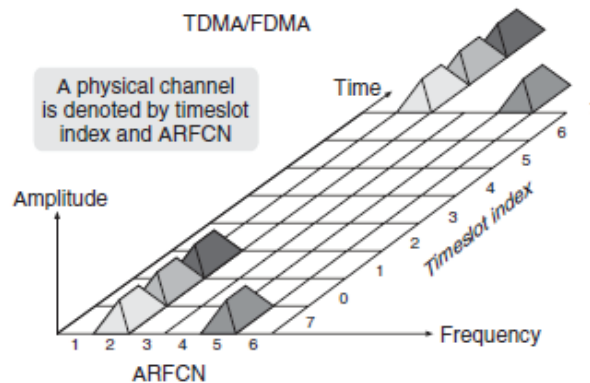


Figure 30.A.2 Time Division Multiple Access/Frequency Division Multiple Access system.

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³ This applies to a GSM900 system, and analogously to other frequency bands.

The important features of the air interface are now described in a step-by-step manner.

Assignment of Timeslots in the Uplink and Downlink

A subscriber utilizes the timeslots with the same number (index) in the uplink and downlink. However, numbering in the uplink is shifted by three slots relative to the numbering in the downlink. This facilitates the design of the UE transmitter/receiver, because reception and transmission do not occur at the same time (compare Figure 30.A.3).

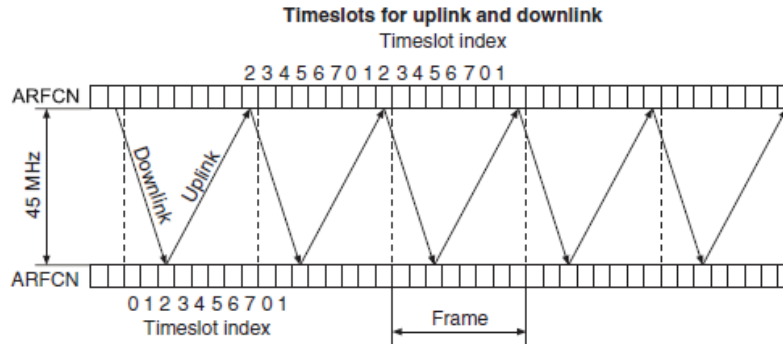


Figure 30.A.3 The alignment of timeslots in the uplinks and downlink.
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Modulation Technique

GSM uses GMSK as a modulation format. GMSK is a variant of Minimum Shift Keying (MSK) where the data sequence is passed through a filter with a Gaussian impulse response (time bandwidth product $B_G T = 0.3$) (see Chapter 10).

This filtering is rather hard. Therefore, the spectrum is rather narrow, but there is a significant amount of Inter Symbol Interference (ISI). On the other hand, the ISI due to delay dispersion of the wireless channel is usually much more severe. Thus, some kind of equalization has to be used anyway. Figure 30.A.4 illustrates a typical example of a phase trellis of this kind of GMSK and of pure MSK for comparison. The detection method is not specified by the standard. Differential detection, coherent detection, or limiter–discriminator detection might be employed.

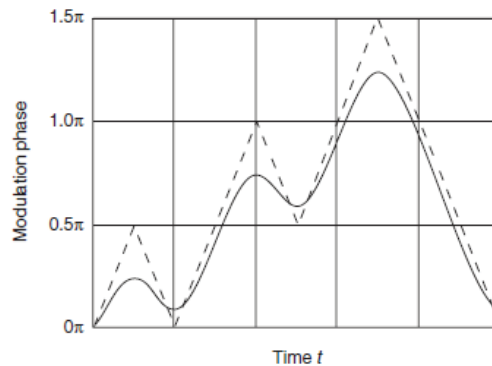


Figure 30.A.4 Phase diagram for the bit sequence 1011011000 for Gaussian minimum shift keying with $B_G T = 0.3$ (solid line) and pure minimum shift keying (dashed line).

Power Ramping

Were a transmitter to start data transmission right at the beginning of each timeslot, it would have to be able to switch on its signal within a very short time (much shorter than a symbol period). Similarly, at the end of a timeslot, it would have to stop transmitting abruptly, so as not to create interference with the next timeslot. This is difficult to implement in hardware and – even if it could be realized – the sharp transition in the time domain would lead to broadening of the emission spectrum. Therefore, GSM defines a time during which signals are smoothly turned off and on (see Figure 30.A.5). In the case when a transmitter is emitting the maximal signal power it has to ramp up from 2×10^{-7} W to 2 W within 28 μ s. During the actual transmission of data, on the other hand, signal power may only deviate by 25% (1 dB) from its nominal value.

Signal Power and Power Control

GSM provides power control for transmission power. Since GSM does not have adaptive modulation and coding, use of more power than necessary to sustain the desired BER is wasteful if energy, and increases adjacent-cell interference (see Chapter 21).

GSM specifies different types of UEs, with different maximum transmission powers though 2-W stations (peak power) are most common. Power control can reduce emitted signal power by about 30 dB; adjustment is done in 2-dB steps. Control is adaptive: the BS periodically informs the UE about the received signal level and the UE uses this information to increase or reduce its transmit power. Maximum power levels of the BSs can vary between 2 and more than 300 W. Also, BSs have a similar power control loop that can decrease output power by some 30 dB.

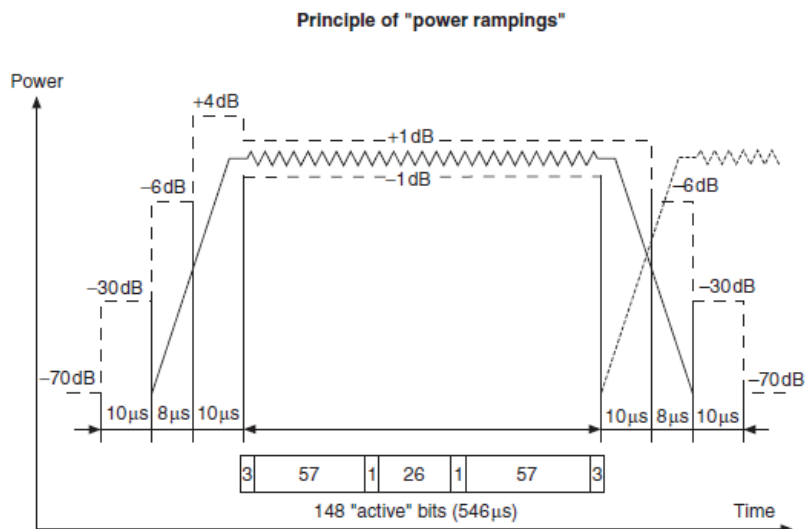


Figure 30.A.5 Power ramping during a timeslot.
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Out-of-Band Emission and Intermodulation Products

The limits for out-of-band emissions are not as severe as, e.g., for analog systems. The maximum permitted out-of-band signal power at both BS and UE is roughly -30 dBm, which is a very high value for wireless communications. However, in the band from 890 to 915 MHz (the uplink band), the power emitted by the BS must not exceed -93 dBm. This is necessary because the BS has to receive signals from UEs, with signal levels as low as -102 dBm, in this band. Furthermore, transmit antennas are located close to receive antennas (or even

colocated) at the BS, and therefore any out-of-band emission in this band causes severe interference.⁴ Similar limits apply for the intermodulation products.⁵

Structure of a Timeslot

Figure 30.A.6 illustrates the data contained in a timeslot with a length of 148 bits. However, not all of these bits are payload data. Payload data are transmitted over two blocks of 57 bits. Between these blocks is the so-called *midamble*. This is a known sequence of 26 bits and provides the training for equalization, which will be covered in Section 30.A.7. Furthermore, the midamble serves as an identifier of the BS. There is an extra control bit between the midamble and each of the two data-containing blocks; the purpose of these control bits are explained in Section 30.A.4. Finally, the transmission *burst* starts and ends with three *tail bits*. These bits are known, and enable termination of Maximum Likelihood Sequence Estimation (MLSE) in defined states at the beginning and end of the detection of burst data. This reduces the complexity and increases the performance of decoding (see also Chapter 13). The timeslots end with a *guard period* of 8.25 bits. Apart from “normal” transmission bursts, there are other kinds of bursts. UEs transmit *access bursts* to establish initial contact with the BS. *Frequency correction bursts* enable frequency correction of the UEs. *Synchronization bursts* allow UEs to synchronize to the frame timing of BSs. These bursts will be explained in more detail in Section 30.A.4.2.

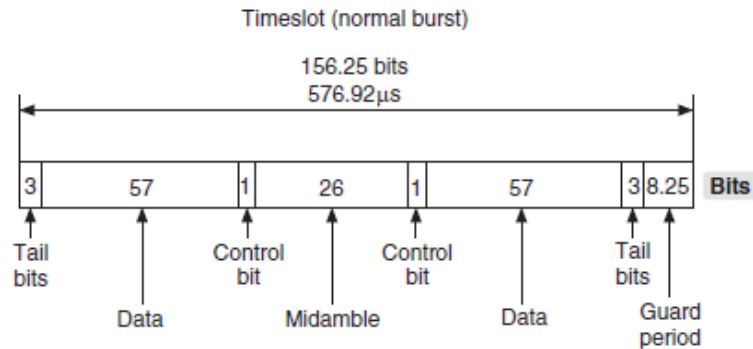


Figure 30.A.6 Functions of the bits of a normal transmission burst.

30.A.4 Logical and Physical Channels

In addition to the actual payload data, GSM also needs to transmit a large amount of signaling information. These different types of data are transmitted via several *logical channels*. The name stems from the fact that each carries a specific type of data (hence “logical”) and is then mapped into specific timeslots that are (parts of) *physical channels*. The first part of this section discusses the kind of data that is transmitted via logical channels. The second part describes the *mapping* of logical channels to physical channels.

30.A.4.1 Logical Channels

Traffic CHannels (TCHs)

Payload data are transmitted via the TCHs. The payload might consist of encoded voice data or “pure” data. There is a certain flexibility regarding the data rate: *Full-rate Traffic CHannels* (TCH/F) and *Half-rate Traffic CHannels* (TCH/H). Two half-rate channels are mapped to the same timeslot, but in alternating frames.

⁴ Similarly, emissions in the bands used by other systems such as UMTS have strict limits.

⁵ Note that intermodulation products can only be found at the BS, as only the BS transmits on several frequencies simultaneously.

Full-Rate Traffic Channels

- Full-rate voice channels: the output data rate of the voice encoder is 13 kbit/s. Channel coding increases the effective transmission rate to 22.8 kbit/s.
- Full-rate data channels: the payload data with data rates of 9.6, 4.8, or 2.4 kbit/s are encoded with Forward Error Correction (FEC) codes and transmitted with an effective data rate of 22.8 kbit/s.

Half-Rate Traffic Channels

- Half-rate voice channels: voice encoding with a data rate as low as 6.5 kbit/s is feasible. Channel coding increases the transmitted data rate to 11.4 kbit/s.
- Half-rate data channels: payload data with rates of 4.8 or 2.4 kbit/s can be encoded with an FEC code, which leads to an effective transmission rate of 11.4 kbit/s.

Broadcast Channels (BCHs)

BCHs are only found in the downlink. They serve as *beacon* signals. They provide the UE with the initial information that is necessary to start the establishment of any kind of connection. The UE uses signals from these channels to establish a synchronization in both time and frequency. Furthermore, these channels contain data regarding, e.g., cell identity. As the BSs are usually not synchronized with respect to each other, the UE has to track these channels not only before a connection is established, but all the time, in order to provide information about possible HOs.

Frequency Correction Channels (FCCHs) The carrier frequencies of the BSs are usually very precise and do not vary in time, as they are based on rubidium clocks. However, dimension considerations and price considerations make it impossible to implement such good oscillators in UEs (compare Chapter 17). Therefore, the BS provides the UE with a frequency reference (an unmodulated carrier with a fixed offset from the nominal carrier frequency) via the FCCH. The UE tunes its carrier frequency to this reference; this ensures that both the UE and the BS use the same carrier frequency.

Synchronization Channel (SCH) In order to transmit and receive bursts appropriately, a UE not only has to be aware of the carrier frequencies used by the BS but also of its frame timing on the selected carrier. This is achieved with the SCH, which informs the UE about the frame number and the *Base Station Identity Code* (BSIC). Decoding of the BSIC ensures that the UE only joins admissible GSM cells and does not attempt to synchronize to signals emitted by other systems in the same band.

Broadcast Control Channel (BCCH) Cell-specific information is transmitted via the BCCH. This includes, e.g., *Location Area Identity* (LAI),⁶ maximum permitted signal power of the UE, actual available TCH, frequencies of the BCCH of neighboring BSs that are permanently observed by the UE to prepare for a handover, etc.

Common Control Channels (CCCHs)

Before a BS can establish a connection to a certain UE, it has to send some signaling information to all UEs in an area, even though only one UE is the desired receiver. This is necessary because in the initial setup stage, there is no *dedicated* channel established between the BS and a UE. CCCHs are intended for transmission of information to all UEs.

Paging Channel (PCH) When a request – e.g., from a landline – arrives at the BS to establish a connection to a specific UE, the BSs within a location area send a signal to all UEs within their range. This signal contains either the permanent *International Mobile Subscriber Identity* (IMSI) or the *Temporary Mobile Subscriber*

⁶ A Location Area (LA) is a set of cells, within which the UE can roam without updating any location information in its HLR.

Identity (TMSI) of the desired UE. The desired UE continues the process of establishing the connection by requesting (via a Random Access CHannel (RACH)) a TCH, as discussed below. The PCH may also be used to broadcast local messages like street traffic information or commercials to all subscribers within a cell. Evidently, the PCH is only found in the downlink.

Random Access CHannel (RACH) A mobile subscriber requests a connection. This might have two reasons. Either the subscriber wants to initiate a connection, or the UE was informed about an incoming connection request via the PCH. The RACH can only be found in the uplink.

Access Grant CHannel (AGCH) Upon the arrival of a connection request via the RACH, the first thing that is established is a Dedicated Control CHannel (DCCH) for this connection. This channel is called the *Standalone Dedicated Control CHannel* (SDCCH), which is discussed below. This channel is assigned to the UE via the AGCH, which can only be found in the downlink.

Dedicated Control Channels (DCCHs)

Similar to the TCHs, the DCCHs are bidirectional – i.e., they can be found in the uplink and downlink. They transmit the signaling information that is necessary during a connection. As the name implies, DCCHs are *dedicated* to one specific connection.

Standalone Dedicated Control CHannel (SDCCH) After acceptance of a connection request, the SDCCH is responsible for further establishing this connection. The SDCCH ensures that the UE and the BS stay connected during the authentication process. After this process has been finished, a TCH is finally assigned for this connection via the SDCCH.

Slow Associated Control CHannel (SACCH) Information regarding the properties of the radio link are transmitted via the SACCH. This information need not be transmitted very often, and therefore the channel is called *slow*. The UE informs the BS about the strength and quality of the signal received from serving BSs and neighboring BSs. The BS sends data about the power control and runtime of the signal from the UE to the BS. The latter is necessary for the *timing advance* (Chapter 18).

Fast Associated Control CHannel (FACCH) The FACCH is used for HOs that are necessary for a short period of time; therefore, the channel has to be able to transmit at a higher rate than the SACCH. Transmitted information is similar to that sent by the SDCCH.

The SACCH is associated with either a TCH or a SDCCH; the FACCH is associated with a TCH.

30.A.4.2 Mapping Between Logical and Physical Channels

The signals of logical channels described above have to be transmitted via physical channels, which are represented by the timeslot number and the ARFCN. In order to better understand the mapping, we first have to realize that the time dimension is not only partitioned into periodically repeated frames of eight timeslots each, but that these frames and timeslots are the smallest units in the time grid. In fact, multiple frames are combined on different levels to make bigger frames (see Figure 30.A.7).

We have already seen above that eight timeslots with a duration of 577 μ s each are combined as a frame. The duration of this frame, 4.61 ms, is the basic period of a GSM system. A total of 26 of these frames are combined as a *multiframe*, which has a duration of 120 ms. Furthermore, 51 of these multiframe are contained in one *superframe*, which has a length of 6.12 s. Finally, 2,048 of these superframes are combined into one *hyperframe*, which lasts 3 h and 28 min. The hyperframe is implemented mainly for cryptographic reasons, in order to guarantee privacy over the air interface. Therefore, encryption is applied to the payload data and the period of the encryption algorithm is exactly the length of one hyperframe.

Understanding the multiple frame structure enables us to discuss which timeslot contains which logical channel. Not all timeslots have to be used for the TCH, as the available data rate on the physical channel is $2 \cdot 57 \text{ bits}/4.615 \text{ ms} = 24.7 \text{ kbit/s}$, while a full rate TCH requires only a 22.8-kbit/s data rate. Therefore, the remaining 1.9 kbit/s may be used for other logical channels.

SACCH As discussed above, 26 frames are combined as a multiframe. Of these 26, only 24 frames are dedicated to the TCH. The 13th (and sometimes the 26th) frame are used by the SACCH. The 26th frame is only employed if two half-rate connections share one physical channel; otherwise the timeslot of the 26th frame is an *idle frame*. The transmission rate of the SACCH is 950 bit/s. The data transmitted via the SACCH is processed differently from the data in the TCH. The bits of four consecutive SACCH bursts are processed together. For this purpose, four multiframes might be combined into a (nameless) higher order frame of length 480 ms. These four SACCH bursts contain 456 bits associated with SACCH data and are used to transmit 184 actual data bits. The data bits are (i) first encoded with a (224, 184) block code, (ii) have four tail bits added, and (iii) then everything is encoded with the regular rate-1/2 convolutional encoder; this leads to the total of $2 \cdot 228 \text{ bit} = 456 \text{ bit}$.

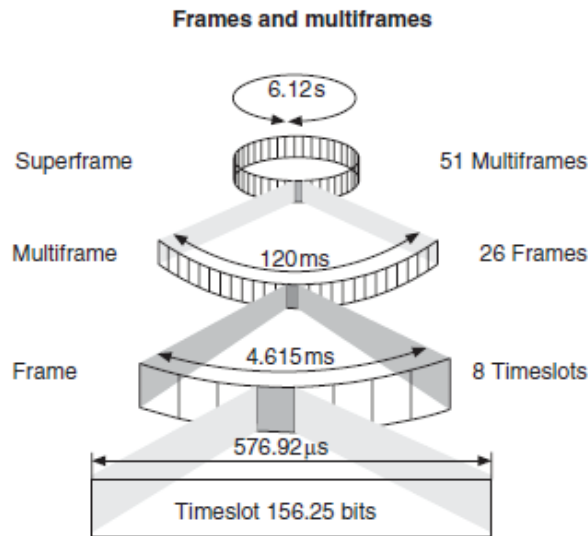


Figure 30.A.7 Structure of Global System for Mobile communications frames for traffic channels.

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FACCH An FACCH does not have to be permanently available. It is only necessary in special situations – e.g., when a handover has to be performed. Therefore, no timeslots are *reserved* for the FACCH. Instead, normal TCH-related bursts of a connection are partly used for FACCH purposes in case this is required. The above-mentioned control bits (stealing bits) between the midamble and the datablocks of a burst indicate whether an FACCH is present in this burst or not – i.e., “steals” bits from the TCH. The 184 bits of an FACCH are encoded in the same way as SACCH bits. In order to transmit the resulting 456 bits via the normal TCH timeslots, eight consecutive frames are used: the even payload bits of the first four bursts and the odd bits of the second four bursts are replaced by bits from the FACCH.

Common Logical Channels The FACCH and SACCH use the physical channel of the associated connection. This is possible as the physical channel supports a slightly higher data rate than is necessary for one TCH connection. Therefore, it is possible to transmit signaling in timeslots belonging to the same physical channel. However, the other logical signaling channels are not associated with a TCH connection, either because they are required for *establishing* a connection or because they are used even in the absence of a TCH channel. Therefore, all these channels operate in the first burst of each frame of the so-called “BCCH carrier.” This assignment strategy makes sure that one physical channel in each cell is permanently occupied. This leads, of course, to a loss of capacity, especially in cells that use only one carrier. However, there is one option to overcome this: if the cell is full, no new connections can be established. Therefore, no timeslots have to be

reserved for signaling related to new connections, and also the first slot of the BCCH carrier can be used for a normal TCH channel.⁷ Furthermore, the frames are combined as higher order frames in a different way. A total of 51 frames are combined into a multiframe, which has a duration of 235 ms. CCCHs are unidirectional, with the RACH being the only channel in the uplink, while several common channels exist in the downlink.

RACH The RACH is necessary only for the uplink. During each multiframe, 8 data bits, encoded into 36 bits, are transmitted via the RACH. These 36 bits are transmitted as an *access burst*. The structure of an access burst has to differ from normally transmitted bursts. At the time the UE requests a connection, it is not yet aware of the runtime of the signal from the UE to the BS. This runtime might be in the range from 0 to 100 μ s where the maximal value is defined by the maximal cell range of 30 km. Therefore, a larger guard time is necessary to ensure that a random burst does not collide with other bursts in adjacent timeslots. After the connection is established, the BS informs the UE about the runtime and therefore the UE can reduce the size of the guard times by employing *timing advance*. A complete random access burst has the following structure. It starts with 8 tail bits, which are followed by 41 synchronization bits. Afterward, the 36 bits of encoded data and 3 additional tail bits are transmitted. This adds to a total of 88 bits and leaves a guard time of 100 μ s at the end, which corresponds to 68.25 bits. As the RACH is the only unassociated control channel in the uplink, the timeslot numbered 0 may be used for random access burst in every frame.

Common Channels in the Downlink The other common channels – such as FCCH, SCH, BCCH, PCH, and AGCH – can only be found in the downlink and have a fixed order in the multiframe. Figure 30.A.8 illustrates this structure. Remember that only timeslot 0 in each frame carries a CCCH. Of the 51 frames in this multiframe, the last one is always idle. The remaining 50 frames are divided into blocks of 10 frames. Each of these blocks starts with a frame containing the FCCH. Afterward the SCH is transmitted during the next frame. The first block of frames contains four BCCHs (in frames 3–6) followed by four frames which contain the PCH or AGCH (frames 7–10). The other four blocks of 10 frames also start with the FCCH and SCH frames, and then consist either of PCH- or AGCH-carrying frames. The FCCH and the SCH employ bursts that have a special structure (this is discussed in the next section). As the UEs of neighboring cells continuously evaluate the signal strength of the first timeslot of the frames on the BCCH carrier, the BS always has to transmit some information during these timeslots, even when there is no connection request.

SDCCH The SDCCH may occupy a physical channel by itself, or – in case the common channels do not occupy all the available slots on the BCCH – it may be transmitted during the first timeslots on the BCCH. In the latter case, either four or eight SDCCHs share this physical channel.

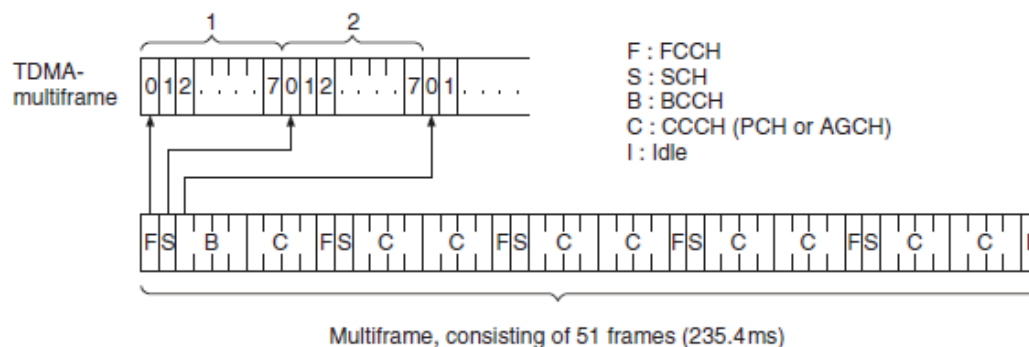


Figure 30.A.8 Mapping of broadcast channels (FCCH, SCH, and BCH) and common control channels to timeslots numbered 0 (compare [CME 20, 1994]).

⁷ Nevertheless, this option is not implemented by most providers.

30.A.5 Synchronization

Up to now, we have assumed that the BS and the UE are synchronized in time and frequency. However, only the BS is required by the standard to have a high-quality time and frequency reference. For the UE, such a reference would be too expensive. Thus, the UE synchronizes its frequency and time references with those of the BS. This is done in three steps: first, the UE tunes its carrier frequency to that of the BS. Next, the UE synchronizes its timing to the BS by using synchronization sequences. Finally, the timing of the UE is additionally shifted with respect to the timing of the BS to compensate for the runtime of the signal between the BS and UE (*timing advance*).

30.A.5.1 Frequency Synchronization

As mentioned before, the BS uses very precise rubidium clocks, or GPS (Global Positioning System) signals, as frequency references. Due to space and cost limitations, the oscillators at the UE are quartz oscillators, which have much lower precision. Fortunately, this is not a problem, since the BS can transmit its high-precision frequency reference periodically and the UE can adjust its local oscillator based on this received reference. Transmission of the reference frequency is done via the FCCH. As we discussed in the previous section, the FCCH is transmitted during timeslots with index 0 of roughly every tenth frame on the BCCH. An FCCH burst consists of 3 tail bits at the beginning, 142 all-zero bits in the middle, and 3 tail bits at the end. The usual guard period (length equivalent to 8.25 bits) is appended. It should be noted that it is not the carrier frequency that is transmitted as a reference, but rather the carrier modulated with a string of zeros. This equals a sinusoidal signal with a frequency that is the carrier frequency *offset* by the MSK modulation frequency. As this offset is completely deterministic, it does not change the principle underlying the synchronization process.

30.A.5.2 Time Synchronization

Time synchronization information is transmitted from the BS to the UE via the SCH. SCH bursts contain information regarding the current index of the hyperframe, superframe, and multiframe. This is not a lot of information, but has to be transmitted very reliably. This explains the relatively complex coding scheme on the SCH. The UE uses the transmitted reference numbers regarding the multiframes, etc. to set its internal counter. This internal counter is not only a time reference with respect to the timeslot and frame grid, but also serves as a time reference within a timeslot with a quarter-bit precision. This reference is initially adjusted by considering the start of SCH bursts received at the UE. The UE then transmits the RACH burst relative to this internal reference. Based on the reception of the RACH, the BS can estimate the roundtrip time between the BS and the UE and use this information for timing advance.

30.A.5.3 Timing Advance

GSM supports cell ranges of up to 30 km, so that propagation delay between the BS and the UE might be as big as 100 μ s. Thus, the following situation might occur: consider user A being at a 30-km distance from the BS, and transmitting bursts in timeslot TS 3 of every frame. User B is located close to the BS and accesses timeslot TS 4. The propagation delay of user A is around 100 μ s, whereas the propagation delay of user B is negligible. Therefore, if propagation delay is not compensated, the end of a burst from user A partly overlaps with the beginning of a burst from user B at the BS (this situation is illustrated in Figure 30.A.9).

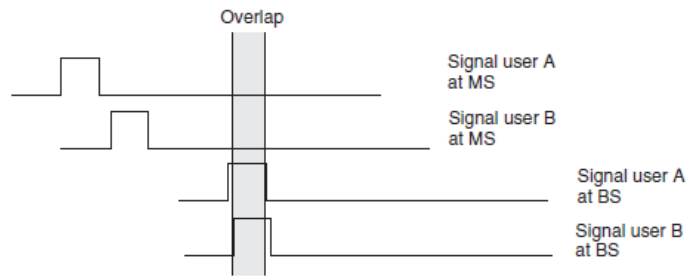


Figure 30.A.9 Overlapping bursts assuming uncompensated propagation delay

To overcome this problem, the propagation delay from UE to BS is estimated by the BS during the initial phase of establishing a connection. The result is transmitted to the UE, which then sends its bursts *advanced* (with respect to the regular timing structure) to ensure that the bursts *arrive* within the dedicated timeslots at the BS. As the access bursts are transmitted before the UE is aware of the propagation delay, it now becomes clear why they must have a bigger guard period than normal transmission bursts: the guard period must be big enough to accommodate the worst case propagation delay – i.e., an UE at the boundary of a cell with maximum size.

There are some very big cells in rural areas where propagation delays exceed foreseen timing advances. In these cases, it might be necessary to use only every second timeslot as otherwise timeslots from different users would collide at the BS. This implies a waste of capacity. However, as this might only occur in rural areas with big cell ranges and low subscriber densities, the actual loss for the provider is small.

30.A.5.4 Summary of Burst Structures

Finally, Figure 30.A.10 provides an overview of the different kinds of bursts and illustrates the functions of their bits.

Normal

3 start bits	58 data bits (encrypted)	26 training bits	58 data bits (encrypted)	3 stop bits	8.25 bits guard period
--------------	--------------------------	------------------	--------------------------	-------------	------------------------

FCCH burst

3 start bits	142 zeros			3 stop bits	8.25 bits guard period
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SCH burst

3 start bits	39 data bits (encrypted)	64 training bits	39 data bits (encrypted)	3 stop bits	8.25 bits guard period
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RACH burst

8 start bits	41 synchronization bits	36 data bits (encrypted)	3 stop bits	68.25 bits extended guard period	
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Dummy burst

3 start bits	58 mixed bits	26 training bits	58 mixed bits	3 stop bits	8.25 bits guard period
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Figure 30.A.10 Structure of timeslots in the Global System for Mobile communications.

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30.A.6 Coding

To transmit speech via the physical GSM channel the “speech signals” have to be translated into digital signals. This process should maintain a certain speech quality while keeping the required data rate as low as possible (see also Chapter 24). Different forms of speech coding were considered for GSM, and finally a *Regular Pulse Excited with Long Term Prediction* (RPE-LTP) solution was chosen (see Chapter 24). The digitized speech that is obtained in such a way then has to be protected by FEC in order to remain intelligible when transmitted over typical cellular channels (uncoded Bit Error Rates (BERs) of $\sim 10^{-3}$ to 10^{-1}). Both block and convolutional codes are used for this purpose in GSM.

Thus, voice transmission in GSM represents a typical example of the paradox of speech communications. First, redundancy is removed from the source data stream during the speech-coding process, and then redundancy is added in the form of error-correcting coding before transmission. The reason for this approach is that the original redundancy of the speech signal is rather inefficient at ensuring intelligibility of speech when transmitted over wireless channels. In this section, we first describe voice encoding, and subsequently channel coding; these can be seen as applications of the principles expounded in Chapters 24 and 13, respectively.

30.A.6.1 Voice Encoding

Like most voice encoders (also referred to as *vocoders*), the GSM vocoder is not a classical source-coding processes like, e.g., the *Huffman code*. Rather, GSM uses a lossy compression method, meaning that the original signal cannot be reconstructed perfectly, but that the compression and decompression procedures lead to a signal which is similar enough to the original one to allow comfortable voice communications. As GSM has evolved, so has the speech coder. For the first release of GSM, an RPE-LTP approach was used. The idea behind this approach is to consider the human voice as output from a time-varying filter bank which is excited periodically. Both parameters describing the filter bank and the excitation process are transmitted. Since the samples of a voice signal are correlated, any sample can be predicted approximately by linearly combining previous samples. Evidently, correlation reflects the redundancy of the voice signal. However, the correlation properties of the signal vary with time, therefore the filter bank has to be time varying as well. Later on, an enhanced speech coder was introduced that improved speech quality without increasing the required data throughput; it is based on Algebraic Code Excited Linear Prediction Coder (ACELP).

The data created by the vocoder are divided into different classes, which have different vulnerabilities to bit errors. By this we mean that the bits have different levels of importance for the *perceived* quality of the reconstructed signal. The bits in class 1a are important, as an error in these bits is perceived as a gross distortion of the signal. Therefore, they are protected by a convolutional code and additional block coding. The slightly less important bits of class 1b are protected just by a convolutional code, while the bits associated with class 2 are transmitted without further channel encoding.

Another method to reduce the data rate is *Voice Activity Detection* (VAD). It detects periods when the user is not speaking, and ceases transmission during these periods – this is *Discontinuous Transmission* (DTX). DTX increases the battery lifetime of the UE and reduces co-channel interference with other users.

30.A.6.2 Forward Error Correction

Let us first give an overview of the encoding procedure. Figure 30.A.11 illustrates the channel coding applied to voice data. For every 20-ms voice signal there are 50 very important bits (class 1a). A block code adds 3 parity bits. This coding is not error *correcting*, but only allows *detection* of bit errors within these 50 bits. The 132 bits of class 1b are attached. After attaching 4 tail bits to determine the final state of the Viterbi decoder, a convolutional code with rate-1/2 is applied. This results in 378 bits which are transmitted together with the 78 bits of class 2. Thus, for every 20 ms of the voice signal, 456 bits have to be transmitted. In the following, the details of the different encoder blocks will be discussed.

Block Encoding

Block Encoding of Voice Data As discussed above, only class-1a bits of the voice data are encoded using a (53,50) block code. This is a very “weak” block code. It is only supposed to detect bit errors and cannot detect

more than three bit errors within the 50 class 1a bits reliably. However, this is sufficient, since a block is completely discarded if an error is detected within the class-1a bits; the receiver then smoothes the resulting signal by “inventing” a block. Figure 30.A.12 shows the linear shift register representation of the block encoder. As the code is systematic, the 50 data bits pass through the encoder unchanged. However, each of them impacts the state of the shift register. The final state of the shift register determines the 3 parity bits which are attached to the 50 class-1a bits. Class 1a, 1b, and parity check bits are then reordered and interleaved. Finally, four all-zero tail bits are attached, which are needed for the convolutional decoder (see below).

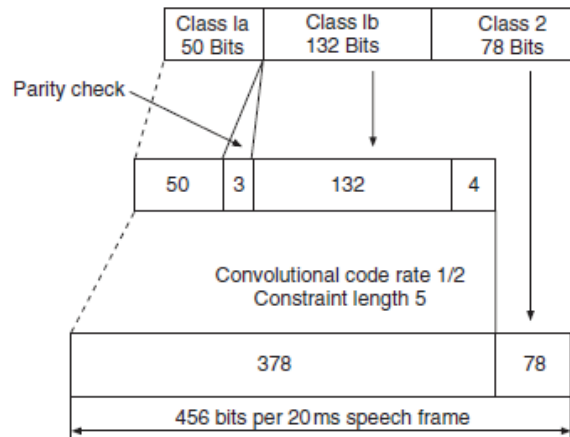


Figure 30.A.11 Channel coding for voice data in the Global System for Mobile communications. Reproduced with permission from Rappaport [1996] © IEEE.

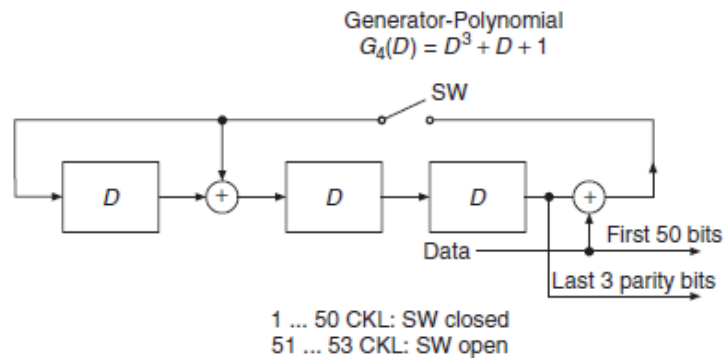


Figure 30.A.12 Shift register structure for voice block encoding, Cla (53,50) systematic, cyclic block encoder. In this figure: CKL, Clock; SW, Switch. Reproduced with permission from Steele and Hanzo [1999] © J. Wiley & Sons, Ltd.

Block Encoding of Signaling Data As mentioned in Section 30.A.4.1, the signaling information has to have stronger protection against bit errors than the voice data. While a bit error in voice-related data might lead to an unintelligible audio signal for 20 ms, a bit error in signaling bits can have a more severe impact – e.g., handover to a wrong cell and therefore loss of connection. Thus, higher redundancy is required, which is achieved by concatenated coding (see Sec. 13.2.5). For most of the control channels, only 184 signal bits are transmitted within 20 ms (instead of 260 for speech). These signaling bits are first encoded with a (224,184) Fire code. The Fire code is defined by the generator polynomial:

$$G(D) = D^{40} + D^{26} + D^{23} + D^{17} + D^3 + 1 \quad (30.A.1)$$

Fire codes are block codes which are particularly capable of correcting burst errors. Burst errors are defined as a series of bit errors, meaning that two or more consecutive bits are wrong; such error bursts occur, e.g., when Viterbi decoding fails (see Chapter 13.3). A total of 4 tail bits are attached to the resulting 224 bits. The result is fed into the convolutional encoder at code rate-1/2, which is the same as that used for class 1 of the voice signal. For selected logical signaling channels, such as RACH and SCH, different generator polynomials are used. The interested reader is referred to [Steele and Hanzo 1999] and the GSM specifications.

Convolutional Encoding

Both the class-1 bits of the voice data and all of the signaling information are encoded with a convolutional coder at code rate-1/2 (see Section 13.3). The bits are fed into a 5-bit shift register. For each new input bit, two codebits are calculated according to the generator polynomials

$$\left. \begin{aligned} G1(D) &= 1 + D + D^3 + D^4 \\ G2(D) &= 1 + D^3 + D^4 \end{aligned} \right\} \quad (30.A.2)$$

and transmitted. The 4 final tail bits attached to the input sequence ensure that the encoder terminates in the all-zero state at the end of each encoded block.

Interleaving

Due to the nature of fading channels, bit errors may occur in bursts in some transmission blocks – e.g., if those blocks were transmitted during a deep fade. Interleaving orders the bits in such a manner that the burst errors due to the channel are (hopefully) distributed evenly (see Section 13.10.1). Evidently, the more the interleaver distributes corrupted bits, the better. However, latency of the speech signal puts an upper limit on *interleaver depth*: In order to give acceptable speech quality, the delay of the signal should be less than 100 ms.

GSM interleaves the data of two blocks (henceforth called “a” and “b”) in the following way: first, each of the blocks is divided into eight subblocks. Specifically, each bit receives an index $i \in \{0, \dots, 455\}$, and the bits are sorted into subblocks with index $k \in \{0, \dots, 7\}$ according to $k = i \bmod 8$. Each subblock of block “a” contributes one half of the bits in a transmission burst (114 bits). The other half is associated with subblocks of either a previous or a succeeding block “b.” Figure 30.A.13 illustrates diagonal interleaving.

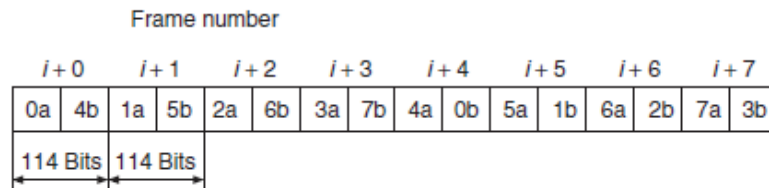


Figure 30.A.13 Diagonal interleaving for traffic channel/slow associated control channel/fast associated control channel data.

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30.A.6.3 Cryptography

One of the most severe shortcomings of analog mobile communications was the ease with which it could be intercepted. Anybody with a frequency scanner was able to eavesdrop on phone conversations. This posed a threat – e.g., for business people dealing with confidential material. Furthermore, even political scandals have been known to develop as a consequence of eavesdropped conversations.

In a digital system, this problem can be solved by “standard” means: once the audio signal is represented by a bitstream, cryptographic procedures, which had long before been developed for military applications, can be easily applied. For GSM, one way of intercepting a conversation is a *man-in-the-middle attack*, which involves implementing a BTS, to which the target UE would log on, and forwarding the intercepted signal (to stop the victim noticing the attack) – a rather cumbersome process. However, security researchers have pointed out serious flaws in the A5/1 cipher that was widely used in GSM, which can be intercepted and decrypted in real time by an adversary. The later-introduced A5/3 cipher has better security, but still flaws exist. Law enforcement typically obtains the collaboration of the network providers, and intercepts conversations not over the air, but *after* the BTS.

Encryption of the transmitted signal is achieved by simply using an XOR operation on the bits on one hand, and a Pseudo Noise (PN)-sequence on the other hand. A PN-sequence is based on feedback linear shift registers and its periodicity is 3.5 hours. Thus, even knowing the sequence does not enable interception, as the listener has to know which part of the sequence is currently in use. The algorithm for encryption of the data, the A5 algorithm, and algorithms involved in authentication, the A3 and A8, were originally only disclosed to members of the MoU. However, they have been reverse engineered and successful attacks have been developed as described above.

30.A.6.4 Frequency Hopping

Slow frequency hopping is an optional feature in GSM, where the carrier frequency is changed for each transmission burst. As outlined in Section 19.1, this helps to mitigate small-scale fading: if the carriers employed are separated by more than one coherence bandwidth of the channel, each frame is transmitted over a channel with independent fading realization. As the data belonging to one payload (voice data) packet are interleaved over eight bursts, the probability that all of them are transmitted via bad channels is negligible. This makes it more likely that the packet can be reconstructed at the receiver. A similar effect occurs for (narrowband) interference.

The coherence bandwidth of GSM channels can vary from several hundred kHz to a few MHz (see App. 7.C). Given that an operator typically owns only a few MHz of spectrum, and only a subset of frequencies can be used in each cell (see Chapter 21), there can be correlation between the frequency channels used for the hopping. Still, frequency hopping provides some advantages even in this case, in particular with respect to interference diversity.

In order for the receiver to follow the hopping pattern of the transmitter, both link ends have to be aware of the order in which carrier frequencies are to be used. The control sequence governing this pattern can specify hops over up to 64 different carriers, but it may also specify the “degenerated” case (no hopping) in which one frequency is used over and over.

The BS determines the actual frequency hopping sequence by selecting one of a set of predefined PN-sequences and matching this sequence to the available frequencies for the cell. Furthermore, it informs the UE about the hopping sequence as well as the phase of the sequence – i.e., when to start – during call setup (for details see [Steele and Hanzo 1999]). Finally, we note that frequency hopping is *not* applied to the physical channels related to BCHs and CCCHs, as the UE is supposed to “find” them easily.

30.A.7 Equalizer

Since the symbol duration of GSM is shorter than typical channel delay spreads, ISI occurs, making it necessary to perform equalization (see Chapter 14). However, as GSM is an open standard, neither the structure nor the algorithms of the equalizer are specified. The signal structure just provides the necessary “hooks” (means of implementation), such as a training sequence used to estimate the channel impulse response. The most important properties of the training sequence are as follows:

- The training sequence is 26 bits long.
- It is transmitted in the middle of a burst and, hence, is also called the *midamble* – it is preceded by 57 data bits, and followed by another 57 data bits.
- Eight different PN-sequences are defined for the midamble – different midambles may be used in different cells, and thus help to distinguish between those cells.

The eight PN-sequences are designed in such a way that their autocorrelation function has a peak of amplitude 26 at zero offset, surrounded by at least five zeros for positive and negative offsets. Therefore, the channel impulse response can be simply estimated by cross-correlating the received midamble with the sequence, as long as the channel impulse response is less than 5 symbol durations long; in Chapter 7 we saw that COST 207 channel models have impulse response lengths of up to 15 μ s, which equals 4 symbol durations.⁸ The cross-correlation thus represents a scaled version of the channel impulse response. This information is used to correct the ISI for all symbols within one burst.⁹

GSM uses a *midamble* for training as it is supposed to support UE speeds of up to 250 km/h. At this speed, the UE covers roughly one-eighth of a wavelength during transmission of a burst (500 μ s). The impulse response of the channel shows some variations over this distance. Were the training sequence transmitted at the beginning of the burst (*preamble*), the resulting channel estimate would no longer be sufficiently accurate at the end of the burst. Since training is transmitted in mid-burst, the estimate is still sufficiently accurate at both the start and the end of a burst.

As mentioned above, the GSM standard does not specify any particular equalizer design. Actually, the equalizer is one of the reasons that products from different manufacturers can differ in price and quality. However, most implemented equalizers are Viterbi equalizers. We also note that Viterbi equalization can be well combined with convolutional decoding.

We emphasize again that a delay-dispersive fading channel with an appropriate equalizer at the receiver leads to *lower* average bit error probabilities than a flat-fading channel since they provide delay diversity (see also Chapter 12).

Table 30.A.1 summarizes the key parameters of GSM.

Table 30.A.1 Key parameters of the Global System for Mobile communications

Parameter	Value
Frequency range	
GSM900	880–915 MHz (uplink) 925–960 MHz (downlink)
GSM1800	1710–1785 MHz (uplink) 1805–1880 MHz (downlink)
Multiple access	FDMA/TDMA/FDD
Selection of physical channel	Fixed channel allocation/intracell handover/frequency hopping
Carrier distance	0.2 MHz
Modulation format	GMSK ($B_G T = 0.3$)
Effective frequency usage per duplex speech connection	50-kHz/channel
Gross bit rate on the air interface	271 kbit/s
Symbol duration	3.7 μ s
Channels per carrier	8 full slots (13 kbit/s user data)
Frame duration	4.6 ms
Maximal RF transmission power at the UE	2 W
Voice encoding	13 kbit/s RPE-LTP
Diversity	Channel coding with interleaving Channel equalization Antenna diversity (optional) Frequency hopping (optional)
Maximal cell range	35 km
Power control	30-dB dynamics

⁸ Note that the constraint length of a Viterbi equalizer can be required to be longer due to other effects – e.g., ISI due to the GMSK. Therefore, 4–6 is a practical value for constraint length.

⁹ Note that the SCH and the RACH use longer training patterns. To simplify implementation, the same algorithm is normally used for equalization of all three different bursts.

30.A.8 Data Transmission – HSCSD, GPRS, and EDGE

When the GSM standard was originally drafted, voice communication was envisioned as the main application. Some data transmission – like the Short Message Service (SMS) and a point-to-point data transmission channel with a 9.6-kbit/s data rate – were already included, but were not considered sufficiently important to merit the introduction of much additional complexity. Thus, data transmission was handled in a circuit-switched mode, just like voice transmission.

In general, the circuit-switched data transmission modes of GSM have severe disadvantages. A main issue is the low data rate of less than 10 kbit/s, though the High Speed Circuit Switched Data (HSCSD) mode provides higher data rates based on circuit-switched transmission. Furthermore, the long time needed to set up a connection, as well as the relatively high costs of holding a connection, make it very unattractive, e.g., for Internet browsing. There was simply a significant mismatch between the low-data-rate connection-based services offered by GSM, and the new Web applications, which require high data volumes in bursts interrupted by long idle periods. Only SMS text messaging proved to be successful. For these reasons, packet-switched (also known as connectionless) transmission (see Section 18.4) was introduced later on.

The first step towards circuit-switched data was GPRS, which provides data rates between 56 and 114 kbit/s. It enables “always-on” internet connections, though the low speed made it mainly attractive for websites with mainly text. The bit rate is partly impacted by the number of timeslots assigned to a user, and partly by the employed coding scheme. GPRS foresees 4 schemes (CS-1 to CS-4), with different amounts of redundancy, that are used depending on the channel quality; it is thus an Adaptive Modulation and Coding (AMC) scheme where the modulation remains fixed (GMSK); the coderates range from $\frac{1}{2}$ to 1.

In contrast to the relatively minor modifications of the PHY, the change from circuit-switched to packet switched connections leads to a significant re-design of the higher layers and in particular the networking structure. The handling of packets is based on similar principles to those of 4G systems, which are described in Chapter 31. We therefore do not discuss them here in detail. The reader interested in the specific details of GPRS can find them in App. 24.B of the second edition of this book, available on the legacy companion website at wiley.com/go/molisch, and [Cai and Goodman 1997] and [Bates 2008].

Further improvement in data rate is provided by Enhanced Data rate for GSM Evolution (EDGE). It uses 8-PSK (in addition to GMSK) as modulation format, thus providing 5 additional MCS schemes, with a peak data rate of 474 kbit/s (assuming use of 8 timeslots for a user). It also introduces incremental-redundancy retransmissions, which have better efficiency than simple retransmissions (see Sec. 13.12). Evolved EDGE, a later update, provides further improvements, including even higher-order modulation (16-QAM and 32-QAM), turbo coding, and use of dual-carriers, all of which results in peak data rates of 1 Mbit/s.

30.A.9 Establishing a Connection and Handover

In this section, we discuss initial establishing of a connection, and the handover procedure, using the logical channels described in Section 30.A.4. Furthermore, we explore the kind of messages that need to be exchanged during these processes and we define various elements of a GSM system required for these functionalities.

30.A.9.1 Identity Numbers

A UE can be localized within the network by using identity numbers. An active GSM UE has multiple identity numbers.

Mobile Station ISDN Number (MS ISDN)

The MS ISDN is the unique phone number of the subscriber in the public telephone network. The MS ISDN consists of *Country Code* (CC), the *National Destination Code* (NDC), which defines the regular GSM provider of the subscriber, and the subscriber number. The MS ISDN should not be longer than 15 digits.

International Mobile Subscriber Identity (IMSI)

The IMSI is another unique identification for the subscriber. In contrast to the MS ISDN, which is used as the phone number of the subscriber within the GSM network *and* the normal public phone network, the IMSI is only used for subscriber identification in the GSM network. It is used by the *Subscriber Identity Module* (SIM), which we explain later, the HLR, and the VLR. It consists again of three parts: the *Mobile Country Code* (MCC, three digits), the *Mobile Network Code* (MNC, two digits), and the *Mobile Subscriber Identification Number* (MSIN, up to ten digits).

Mobile Station Roaming Number (MSRN)

The MSRN is a temporary identification that is associated with a UE if it is not in the area of its HLR. This number is then used for routing of connections. The number consists again of a CC, MNC, and a TMSI, which is given to the subscriber by the GSM network they is roaming into.

International Mobile Station Equipment Identity (IMEI)

The IMEI is a means of identifying hardware – i.e., the actual mobile device. Let us note here that the three identity numbers described above are all either permanently or temporarily associated with the subscriber. In contrast, the IMEI identifies the actual UE used. It consists of 15 digits: six are used for the *Type Approval Code* (TAC), which is specified by a central GSM entity; two are used as the *Final Assembly Code* (FAC), which represents the manufacturer; and six are used as a *Serial Number* (SN), which identifies every UE uniquely for a given TAC and FAC.

30.A.9.2 Identification of a Mobile Subscriber

In analog wireless networks, every UE was uniquely identified by a single number that was permanently associated with it. All connections that were established from this UE were billed to its registered owner. GSM is more flexible in this respect. The subscriber is identified by their SIM, which is a plug-in chipcard roughly the size of a postage stamp (micro- and nano-SIMs, which are popular nowadays, are even smaller; while electronic SIM cards are mainly used for IoT devices and cutting-edge consumer devices that usually do not include GSM phones). A GSM UE can only make and receive calls when such a SIM is plugged in and activated.¹⁰ All calls that are made from the UE are billed to the subscriber whose SIM is plugged in. Furthermore, the UE only receives calls going to the number of the SIM owner. This makes it possible for the subscriber to easily replace the UE, or even rent one for a short time.

As the SIM is of fundamental importance for billing procedures, it has to have several security mechanisms. The following information is saved on it:

- *Permanent security information*: this is defined when the subscriber signs a contract with the operator. It consists of the IMSI, the authentication key, and the access rights.
- *Temporary network information*: this includes the TMSI, location area, etc.
- *Information related to the user profile*: e.g., the subscriber can store their personal phonebook on the SIM – in this way the phonebook is always available, independent of the UE the subscriber uses.

The SIM can be locked by the user. It is unlocked by entering the *Personal Unblocking Key* (PUK). If a wrong code is entered ten times, the SIM is finally blocked and cannot be reactivated. Removing the SIM and then plugging it into the same or another UE does not reset the number of wrong trials. This blocking mechanism is an important security feature in case of theft.

The *Personal Identification Number* (PIN) serves a similar function as the PUK. The user may activate the PIN function, so that the SIM requests a four-digit key every time an UE is switched on. In contrast to the PUK, the PIN may be altered by the user. If a wrong PIN is entered three times, the SIM is locked and may be unlocked only by entering the PUK.

¹⁰ Emergency calls can be made without a SIM.

30.A.9.3 Examples for Establishment of a Connection

In the following, we give two examples for the steps that are performed when a connection is established. Both the user identification numbers and the different logical channels (see Section 30.A.4) play a fundamental role in this procedure.

If a subscriber wants to establish a connection from their UE, the following procedure is performed between the UE and the BTS to initialize the connection:

1. The UE requests an SDCCH from the BS by using the RACH.
2. The BS grants the UE access to an SDCCH via the AGCH.
3. The UE uses the SDCCH to send a request for connection to the MSC. This includes the following activities: the UE tells the MSC which number it wants to call. The authentication is performed; in this context it is evaluated if the UE is allowed to make a requested call (e.g., an international call). Furthermore, the MSC marks the UE as busy.
4. The MSC orders the BSC to associate a free TCH with the connection. The BTS and the UE are informed of the timeslot and carrier number of the TCH.
5. The MSC establishes a connection to the network to which the call should go – e.g., the PSTN. If the called subscriber is available and answers the call, the connection is established.

A call that is incoming from another network starts the following procedure:

1. A user of the public phone network calls a mobile subscriber, or more precisely, a UE ISDN. The network recognizes that the called number belongs to a GSM subscriber of a specific provider, since the NDC in the UE ISDN contains information about the network. The PSTN thus establishes a connection to a gateway MSC¹¹ of the GSM provider.
2. The gateway MSC looks in the HLR for the subscriber's information and the routing information (the current location area of the subscriber).
3. The HLR translates the UE ISDN into the IMSI. If call forwarding is activated – e.g., to a voicemail box – the process is altered appropriately.¹²
4. If the UE is roaming, the HLR is aware of the MSC it is connected to, and sends a request for the MSRN to the MSC that is currently hosting the UE.
5. The hosting MSC sends the MSRN to the HLR. The gateway MSC can now access this information at the HLR.
6. As the MSRN contains an identification number of the hosting MSC, the gateway can now forward the call to the hosting MSC. Additional information – e.g., the caller ID – is included.
7. The hosting MSC is aware of the *location area* of the UE. The location area is the area controlled by one BSC. The MSC contacts this BSC and requests it to page the UE.
8. The BSC sends a paging request to all the BTSs that cover the location area. These transmit the paging information via the BCH.
9. The called UE recognizes the paging information and sends its request for an SDCCH.
10. The BSC grants access to an SDCCH via the AGCH.
11. Establishment of the connection via the SDCCH follows the same steps as described in bullets 3 and 4 of the “UE-initiated call.” If the mobile subscriber answers the incoming call, the connection is established.

30.A.9.4 Examples of Different Kinds of Handovers

A *handover* is defined as the procedure where an active UE switches the BTS with which it maintains a link; it is a vital part of mobility in cellular communications. Handover is performed when another BTS is capable of providing better link quality than the current one. In order to determine whether another BTS could provide better link quality, the UE monitors the signal strength of the BCH of neighboring BTSs. Since the BCH does not use power control, the UE measures the maximum signal power available from other BTSs. It transmits the results of these measurements to the BSC. Furthermore, the currently active BTS measures the quality of the

¹¹ A gateway MSC is an MSC with a connection to the regular phone network.

¹² This information is found in the HLR as well.

uplink and sends this information to the BSC. Based on all this information, the BSC decides if and when to initiate a handover. Since the UE contributes to the handover decision, this procedure is called *Mobile Assisted Hand Over* (MAHO).

Let us now consider some more details in this procedure:

- The received signal strength from different BTSs is averaged over a few seconds (the exact values are selected by the network provider); this ensures that small-scale fading does not lead to a handover. Otherwise, a UE exposed to a similar signal strength from two BTSs would constantly switch from one BTS to the other by just moving over a small distance.
- Receive power is measured at 1-dB resolution in the range of -103 dBm to -41 dBm. The lower bound reflects the sensitivity of GSM receivers – i.e., the minimum signal power required for communication.
- Furthermore, a handover is initialized when the necessary timing advance exceeds the specified limit of 235 μ s. If the UE is so far away from the BTS that a bigger timing advance is necessary, a handover is made to a closer BTS.
- Even more importantly, a handover is initialized when the signal quality becomes too low due to interference.
- The BS transmits (via the BCCH) several parameters that support the handover procedure.

In the following, we describe three different types of HOs: the simplest involves only BTSs controlled by the same BSC. A more complex case arises if two different BSCs are connected to the same MSC. The most complex case involves different MSCs.

Case 30.A.1 – Handover between BTSs Belonging to the same BSC

The steps for this case are illustrated in Figure 30.A.14:

1. The BSC orders the new BTS to activate a new physical channel.
2. The BSC uses the FACCH of the link between the UE and the old BTS in order to transmit information about the carrier frequency and timeslot of the physical channel for the new BTS.
3. The UE switches to the new carrier frequency and timeslot and sends handover access bursts. These bursts are similar to RACH bursts: they are shorter than normal transmission bursts, as the necessary timing advance is unknown and has to be evaluated first by the new BTS.
4. After the new BTS has detected the handover bursts, it sends the necessary timing advance and power control information to the UE via the FACCH of the new channel.
5. The UE informs the BSC that the handover was successful.
6. The BSC requests the old BTS to switch off the old channel.

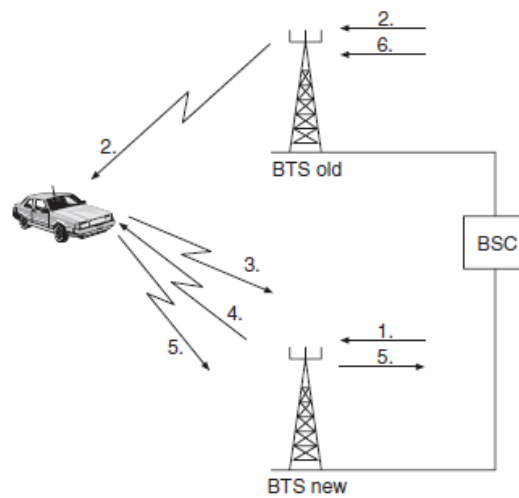


Figure 30.A.14 Handover between two base transceiver stations of the same base station controller.

Case 30.A.2 – Handover between Two BTSs that are Controlled by Different BSCs but the Same MSC

The steps for this case are illustrated in Figure 30.A.15.

1. The old BSC informs the MSC that a handover to a specific BTS is necessary.
2. The old MSC knows which BSC controls this BTS (the new BTS) and informs this new BSC about the upcoming handover.
3. The new BSC requests the new BTS to activate a physical channel.
4. The new BSC informs the UE about the carrier frequency and timeslot for the new link. This information goes via the MSC, the old BSC, and the old BTS, and is finally transmitted to the UE on the FACCH of the old BTS.
5. The UE switches to this new carrier frequency and timeslot and transmits access bursts (compare item 3 in Case 30.A.1).
6. After detecting handover bursts, the BTS transmits information regarding timing advance and power control to the UE via the FACCH.
7. The UE informs the old BSC about the successful handover (via the new BSC and the MSC).
8. The new BSC instructs the old BSC (via the MSC) to relinquish the connection to the UE.
9. The old BSC instructs the old BTS to deactivate the old physical channel.

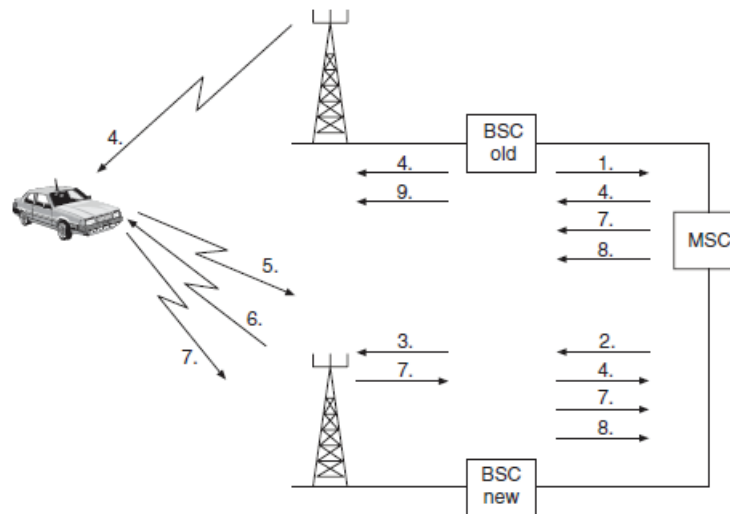


Figure 30.A.15 Handover between two cells belonging to different base station controllers but the same mobile switching center.

Case 30.A.3 – Handover between Two BTSs which are Associated with Two Different MSCs

The steps for this case are illustrated in Figure 30.A.16:

1. The old BSC informs its own MSC (in the following called “MSC-A”) about the necessary handover.
2. MSC-A recognizes that the requested handover involves a BTS associated with another MSC (in the following called “MSC-B”) and contacts this MSC-B.
3. MSC-B associates a handover number with the process, so that it is able to reroute the connection. Subsequently, it informs the new BSC about the upcoming handover.
4. The new BSC orders the new BTS to activate a physical channel.
5. MSC-B gets the information about the carrier frequency and timeslot of the new physical channel and forwards this information to MSC-A. Furthermore, it informs MSC-A about the handover number of the connection.
6. A connection between MSC-A and MSC-B is established.
7. MSC-A informs the UE about the carrier frequency and timeslot of the new physical channel. The information goes from MSC-A via the old BSC and the old BTS, whence it is transmitted to the UE via the FACCH.
8. As in Cases 30.A.1 and 30.A.2, the UE transmits handover bursts on the new physical channel.
9. After detecting the handover bursts, the new BTS instructs the UE about power control and timing advance.
10. The UE informs MSC-A about the successful handover; this information goes via the new link over the new BTS, the new BSC and MSC-B. From this time on, MSC-A forwards the connection to MSC-B. However, the connection is still maintained by MSC-A. MSC-A acts therefore as a so-called *anchor MSC*.
11. The old physical channel is deactivated.
12. After the connection ends, the new location area of the UE is established. Therefore, the VLR of MSC-B informs the HLR that the UE is now in its area and the HLR updates its entries regarding the location of the UE. Furthermore, the HLR request the VLR of MSC-A to delete all entries associated with the UE.

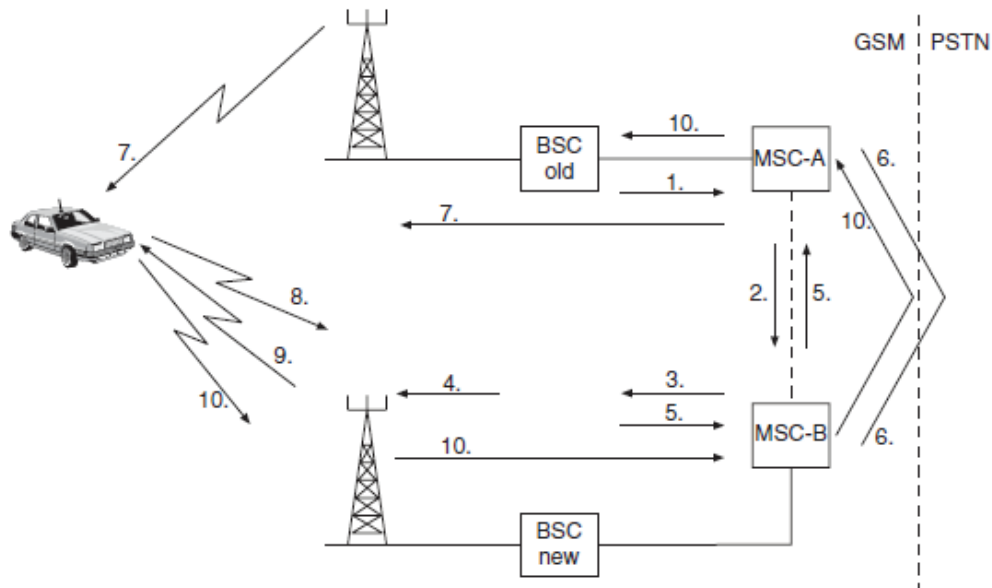


Figure 30.A.16 Handover between two cells belonging to two different mobile switching centers.

We see from these examples that the handover procedure depends on where in the network the switching centers are located.

30.A.10 Services and Billing

30.A.10.1 Available Services

In contrast to analog cellular networks, GSM offers a variety of services in addition to regular phone calls. Although providing those services is not a big effort for the network provider, they were a major motivation for customers to switch from analog systems to GSM or another digital mobile phone system. We thus briefly discuss in the following the services offered by GSM, distinguishing between (i) *teleservices*, (ii) *bearer services*, and (iii) *supplementary services*. Teleservices provide a connection between two communication partners, though they may use some additional devices to exchange information via this connection. Bearer services allow the subscriber to access another system. Thus, they provide a connection from the UE to an access point to another network (not a specific terminal in another network). Supplementary services support or manage both teleservices and bearer services.

Teleservices

- *Regular phone calls*: this is still the most common application of GSM.
- *Dual Tone Multi Frequency (DTMF)*: this is a signaling method to allow the subscriber to control a device connected to a phoneline using the keypad of the UE. A typical example is the remote checking of an answering machine connected to a regular landline.
- *Emergency calls*: calls to emergency numbers have priority in GSM. If a cell is already full, an emergency call leads to interruption of another connection. Remember also that emergency calls can be made from any UE, even without a valid SIM.
- *FAX*: the protocol that is used to transmit *Commite' Consultatif International de Telegraphique et Telephonique (CCITT) group 3* faxes is incompatible with regular GSM connections. The classical fax protocol was developed to transmit a picture that was converted into a specific digital form over an analog

phoneline. GSM, on the other hand, provides either voice connections with a voice-encoding function or data transmission channels for digital data. To send or receive a fax via GSM, adapters are thus required both at the GSM network end and at the UE. The *Terminal Adapter Function* (TAF) provides a general interface between GSM devices and other devices, and is combined with a special fax adapter.

- *SMS*: a short message consists of up to 160 alphanumeric characters. A short message may be transmitted from or to a UE. If it is transmitted to a UE that is switched off, the message is accepted by the GSM system and stored in a dedicated database (*SMS center*). The message is delivered once the UE is switched on and its location is known. This service has turned out to be staggeringly successful. While today's text messaging services often use different underlying technologies, the principles (and often even the character limitations) of SMS have been retained.
- *Cell broadcast*: a cell broadcast transmits a broadcast message of up to 93 alphanumeric characters to all the UEs within the range of a BTS. This may be used to transmit, e.g., news regarding the local (car) traffic situation.
- *Voicemail*: the voicemail service allows the subscriber to forward incoming calls to a service center of the GSM network that acts like an answering machine. After listening to a message from the subscriber the calling party may leave a voice message, which the subscriber can retrieve by connecting to this service center. Typically, a subscriber is notified about new messages via SMS.
- *Fax mail*: this is a service that allows the subscriber to forward incoming faxes to a special service center in the GSM network. The faxes can be retrieved by the subscriber by connecting to this center either with a GSM connection or from the PSTN.

Bearer Services

- *Connections to the PSTN*: this allows the user to connect, e.g., to modems connected to an analog landline.
- *Connections to the Integrated Services Digital Network (ISDN)*: all digital information can be transmitted over the digital network.
- *Connections to the packet-switched networks*: the user may also access packet-switched networks. This is nowadays the most important of the bearer services.

Supplementary Services

GSM enables the network provider to offer the subscriber a variety of additional, supplementary services, e.g.,

- *Call forwarding*: the user can select under which conditions calls to their mobile number are forwarded to another number: (i) always; (ii) in case the UE cannot be reached; (iii) the user is making or receiving another call; or (iv) the user does not answer after a specified number of rings.
- *Blocking of outgoing calls*: the user or the provider may block outgoing calls under some of the following conditions: (i) all outgoing calls; (ii) all international calls; or (iii) all international calls, with the exception of those to the country of origin in case the subscriber is abroad.
- *Blocking of incoming calls*: this feature is of interest when the subscriber has to pay part of (or all of) the charges for an incoming call. The feature may be activated always or when the user is roaming out of the original network.
- *Advice of charges*: the user may be able to access an estimation of the call charges.
- *Call hold*: the user may put a connection on hold to make or receive another call and then continue with the first connection.
- *Call waiting*: during a call, the subscriber may be informed about another incoming call. They may either answer this call by putting the other call on hold or reject it. This feature is available for all circuit-switched connections except emergency calls.
- *Conference calls*: this feature enables connection to multiple subscribers simultaneously. It is only possible for normal voice communications.
- *Caller ID*: the phone number of the incoming call is displayed.
- *Closed groups*: subscribers in GSM, ISDN, and other networks may be defined as a specific user group. Members of this group can, e.g., be allowed to make calls only within the group.

30.A.10.2 Billing

In GSM, billing for the variety of different subscriber plans is not only an economics issue but also a technical challenge that involves the design of an *OSS*. In contrast to the regular public phone system, not all fees have to be paid by the party initiating the calls. Furthermore, accounting for supplementary services has to be done separately. To give an impression of the complexity of the accounting involved, we discuss one particular example here.¹³

Example 30.A.1 *Billing in the Global System for Mobile communications.*

The example involves the following communication parties:

- Subscriber A originates from Austria but is temporarily in Poland.
- Subscriber B is an English subscriber staying in France with a rented UE but their own English SIM.
- Subscriber C is Italian. They is on vacation and the option “If user does not answer, forward call to Subscriber B” is activated.
- Subscriber D is a subscriber to a U.S. service, but is currently in Mexico.

Communication now follows the steps below:

1. Subscriber A calls subscriber C in Italy.
2. As subscriber C is not answering, the call is forwarded to subscriber B.
3. Subscriber B is in France and is currently speaking on their UE. Therefore, subscriber A activates the option “automatic call to busy UE.” Thus, the UE automatically initiates a call the moment the other UE is no longer busy.
4. After subscriber B finishes their conversation, the UE of subscriber A initiates a connection to the UE of subscriber B.
5. This connection is first routed to England, where the HLR of subscriber B is located.
6. From there it gets forwarded to France, where subscriber B is right now.
7. During the conversation, subscriber B needs some information from subscriber D. Therefore, D initiates a “conference call” and calls subscriber D.
8. The call to subscriber D is first routed to the U.S.A. and from there to Mexico, where subscriber D is temporarily staying.

Now the question arises as to which subscriber is charged for which fees?

- Subscriber A has to pay the fees for a call from Poland to Italy. They has to pay both the “international call” charges, and the roaming fees (as they is not in their home country). Furthermore, A has to pay for the service “automatic call to busy UE.”
- Subscriber B has to pay for a connection from England to France (for the incoming call), the charges for an international call (from France to the U.S.A.), and the roaming charges (initiating a call while being in a different network). Further, B has to pay for the “conference call” feature.
- Subscriber C has to pay for the connection from Italy to England and the “call forwarding” feature involved.
- Subscriber D has to pay for a “received call” in the U.S. (note that in the U.S., the called party pays for a received call the same way as for an active call), and the roaming fees from the U.S. to Mexico.

We see that for the same conversation different subscribers get charged different fees depending on their roaming. Subscribers do not have to be actively involved in the conversation to be charged (see subscriber C in the above example). This example gives a taste of the complexity of the billing software in the *OSS*.

¹³ Note that we base the discussion of this example partly on European billing procedures. This is fundamentally different from U.S. billing. In the U. S., cellular numbers are similar to landline numbers. Therefore, the calling party just pays the regular fees for a landline call, whereas the mobile subscriber pays the landline-to-mobile fees even for incoming calls. However, note that there are very few operative GSM networks remain in the U. S. (though the same billing procedures apply to 4G and 5G networks).

30.A.10 Glossary for GSM

AB	Access Burst
AC	Administration Centre
ACCH	Associated Control CHannel
ACM	Address Complete Message
AGCH	Access Grant CHannel
ARFCN	Absolute Radio Frequency Channel Number
AUC	Authentication Center
BCC	Base station Color Code
BCCH	Broadcast Control CHannel
BCF	Base Control Function
BCH	Broadcast CHannel
Bm	Traffic channel for full-rate voice coder
BN	Bit Number
BNHO	Barring all outgoing calls except those to Home PLMN
BS	Base Station
BSC	Base Station Controller
BSI	Base Station Interface
BSIC	Base Station Identity Code
BSS	Base Station System
BSSAP	Base Station Application Part
BTS	Base Transceiver Station
CA	Cell Allocation
CBCH	Cell Broadcast CHannel
CC	Country Code
CCBS	Completion of Calls to Busy Subscribers
CCCH	Common Control CHannel
CCPE	Control Channel Protocol Entity
CI	Cell Identify
CM	Connection Management
CONP	Connect Number Identification Presentation
CUG	Closed User Group
DB	Dummy Burst
DCCH	Dedicated Control CHannel
DRM	Discontinuous Reception Mechanisms
DTAP	Direct Transfer Application Part
DTE	Data Terminal Equipment
DTMF	Dual Tone Multi Frequency (signalling)
DRX	Discontonuous Reception
DTX	Discontonuous Transmission Mechanisms
EIR	Equipment Identify Register
FB	Frequency correction Burst
FACCH	Fast ACCH
FACCH/F	Full-rate FACCH
FACCH/H	Half-rate FACCH
FCH	Frequency Correction Channel
FN	Frame Number
GMSC	Gateway Mobile Services Switching Centre
GSM	Global System for Mobile communications
HDLC	High Level Data Link Control
HLR	Home Location Register

HMSC	Home Mobile-services Switching Centre
HSN	Hop Sequence Number
IAM	Initial Address Message
ICB	Incoming Calls Barred
ID	Identification
IMEI	International Mobile station Equipment Identity
IMSI	International Mobile Subscriber Identity
ISDN	Integrated Services Digital Network
IWF	Inter Working Function
Kc	Cipher Key
Ki	Key used to calculate SRES
Kl	Location Key
Ks	Session Key
LAC	Location Area Code
LAI	Location Area Identify
LAP	hyphen;Dm Link Access Protocol on Dm Channel
LPC	Linear Prediction Coding (Voice Codec)
LR	Location Register
MA	Mobile Allocation
MACN	Mobile Allocation Channel Number
MAF	Mobile Additional Function
MAIO	Mobile Allocation Index Offset
MAP	Mobile Application Part
MCC	Mobile Country Code
ME	Maintenace Entity
MEF	Maintenace Entity Function
MIC	Mobile Interface Controller
MNC	Mobile Network Code
MS	Mobile Station
MSC	Mobile-services Switching Centre
MSCU	Mobile Station Control Unit
MS	ISDN Mobile Station ISDN Number
MSL	Main Signaling Link
MSRN	Mobile Station Roaming Number
MT	Mobile Terminal
MTP	Message Transfer Part
MUMS	Multi User Mobile Station
NB	Normal Burst
NBIN	A parameter in the hopping sequence
NCELL	Neighbouring (adjacent) Cell
NDC	National Destination Code
NF	Network Function
NM	Network Management
NMC	Network Management Centre
NMSI	National Mobile Station Identification number
NSAP	Network Service Access Point
NT	Network Termination
OACSU	Off Air Call Set Up
O&M	Operations & Maintenance
OCB	Outgoing Calls Barred
OMC	Operations & Maintenance Center
OS	Operating Systems

PAD	Packet Assembly/Disassembly facility
PCH	Paging CHannel
PIN	Personal Identification Number
PLMN	Public Land Mobile Network
PSPDN	Public Switched Public Data Network
PSTN	Public Switched Telephone Network
PTO	Public Telecommunications Operators
RA	Random Mode Request information field
RAB	Random Access Burst
RACH	Random Access Channel
RFC	Radio Frequency Channel
RFN	Reduced TDMA Frame Number
RLP	Radio Link Protocol
RNTABLE	Table of 128 integers in the hopping sequence
RPE	Regular Pulse Excitation (Voice Codec)
RXLEV	Received Signal Level
RXQUAL	Received Signal Quality
SABM	Set Asynchronous Balanced Mode
SACCH	Slow Associated Control Channel
SACCH/C4	Slow, SACCH/C4 Associated Control CHannel
SACCH/C8	Slow, SACCH/C8 Associated Control CHannel
SACCH/T	Slow, TCH Associated Control CHannel
SACCH/TF	Slow, TCH/F Associated Control CHannel
SACCH/TH	Slow, TCH/H Associated Control CHannel
SAP	Service Access Points
SAPI	Service Access Points Indicator
SB	Synchronization Burst
SCCP	Signalling Connection Control Part
SCH	Synchronisation CHannel
SCN	Sub Channel Number
SDCCH	Standalone Dedicated Control CHannel
SDCCH/4	Standalone Dedicated Control CHannel/4
SDCCH/8	Standalone Dedicated Control CHannel/8

Further Reading

This current chapter is of course only a brief overview of GSM technology. Much more detailed information may be found in the GSM specifications. Note, however, that they encompass 5,000 pages and are also written as a technical specification and not as a textbook, and most engineers only read small sections of them. Another useful source of information is the monograph by [Mouly and Pautet 1992]. A detailed GSM chapter can also be found in [Steele and Hanzo 1999], [Steele et al. 2001], and [Schiller 2003]. GPRS is discussed in [Cai and Goodman 1997]; [Bates 2008]; GSM network aspects are discussed in [Eberspaecher et al. 2009].

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